

ECON 100B: Macroeconomics

Class 4: A Model of Production 1 (Jones Chapter 4)

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Agenda

- Announcements
- Wrap-up Chapter 3, An Overview of Long-Run Economic Growth
- Chapter 4: A Model of Production

Outline of Macroeconomics

- Chapter 2 shows how we measure income, $Y = C + I + G + NX$
- **Chapters 3–6 discuss long-run growth**
- Chapters 7–8 talk about inflation and unemployment
- Chapters 9–14 discuss short-run fluctuations, fiscal policy, monetary policy, the Phillips Curve, the Fed
- Chapters 16 & 17 focus on Consumption and Investment
- Chapter 18 looks at Government
- Chapters 19 & 20 address international trade and finance, exchange rates

Last topic from Chapter 3: The ratio scale

Equivalent to a logarithmic scale

Plotting on a ratio scale (a.k.a. log scale)

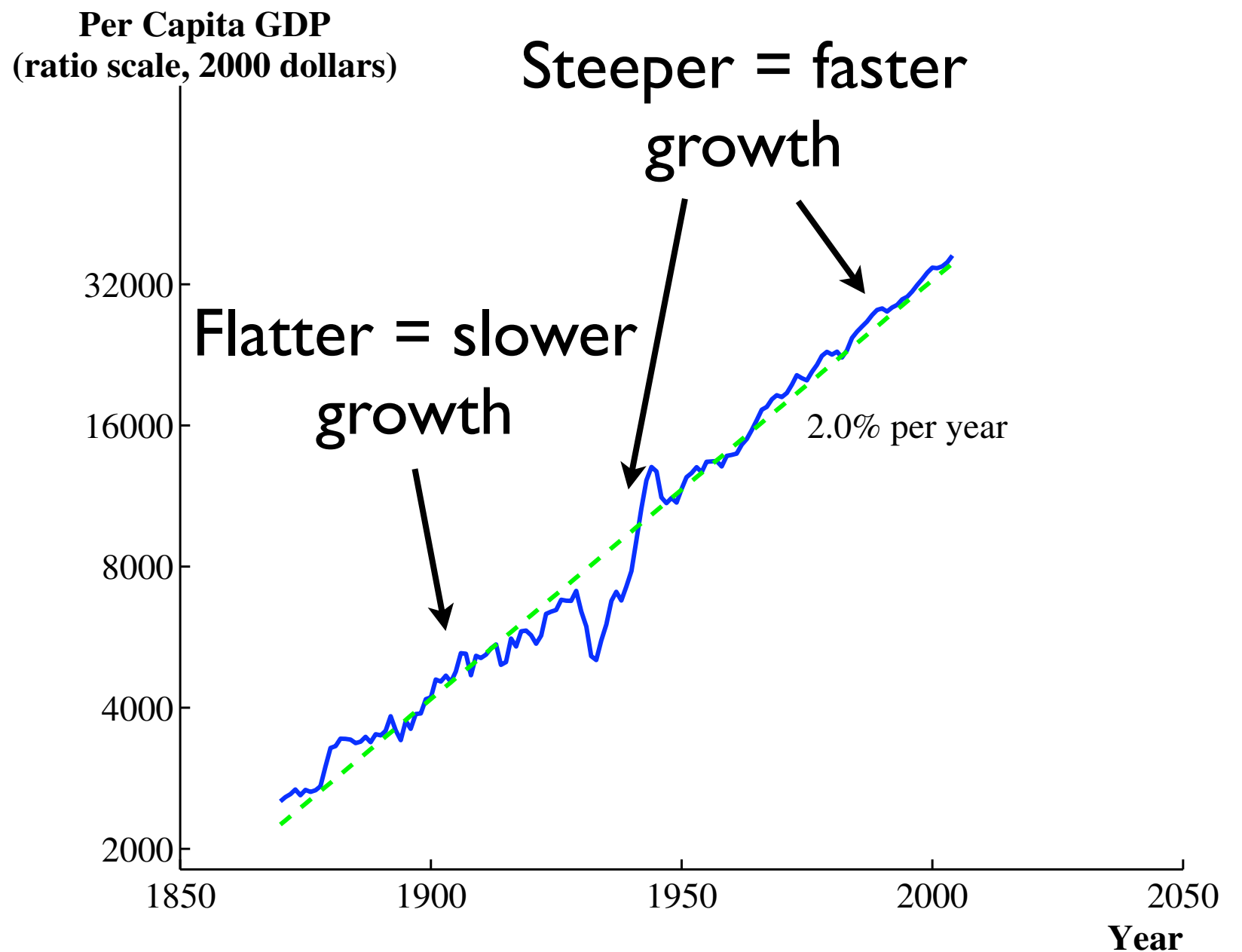
- If the y -axis is scaled in terms of ratios or multiples of an amount rather than its levels,
- Then a series that grows at a constant rate...
appears as a straight line on a ratio scale

On a ratio scale,
equal spacings
are constant
ratios (here, 2:1
or doubling)

GDP per person
has grown at a
fairly constant
rate of 2%

The slopes reveal
faster or slower
growth

Figure 3.5: Per Capita GDP in the United States: Ratio Scale



Note: This is the same data shown in Figure 3.2, but plotted using a *ratio scale*. Notice that the ratios of the equally-spaced labels on the vertical axis are all the same, in this case equal to 2. The dashed line exhibits constant growth at a rate of 2.0 percent per year.

Ratio scales allow us to see and tell stories about shifting growth rates much easier

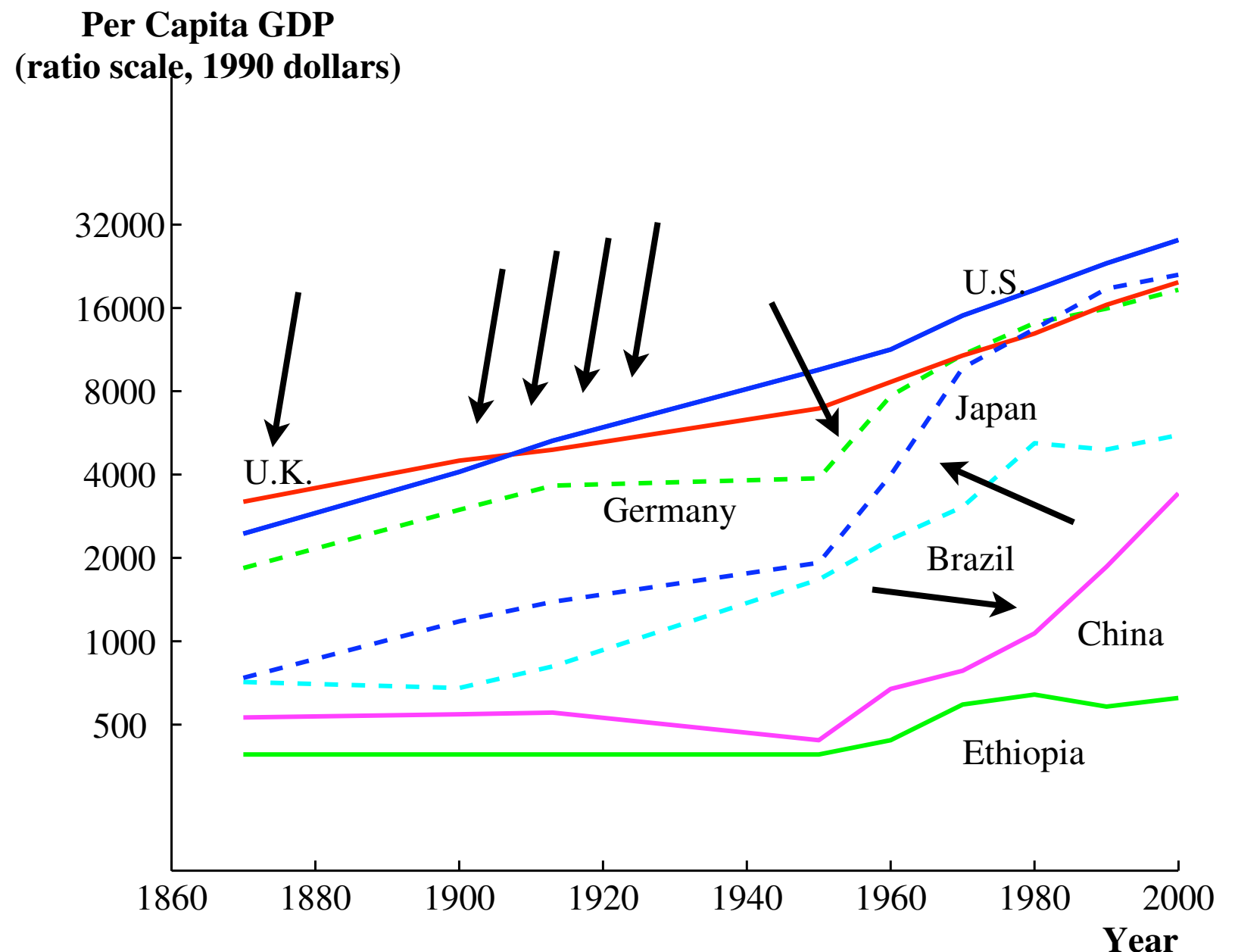
In 1870, the UK was the richest country

But the U.S. grew more rapidly!

Postwar Germany and Japan caught up

China is growing fast!

Figure 3.6: Per Capita GDP since 1870



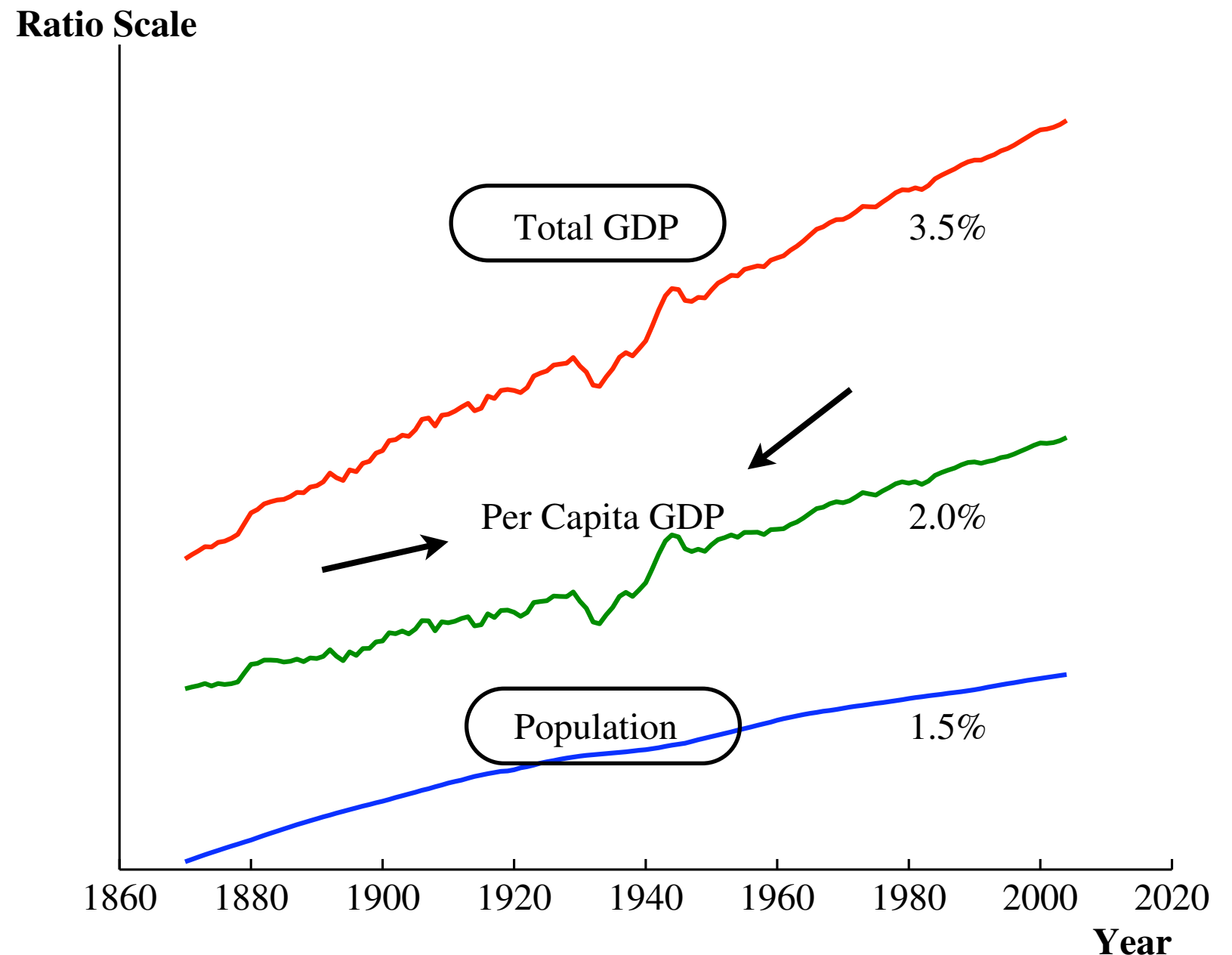
Note: Data from Angus Maddison, *The World Economy: Historical Statistics* (Paris: OECD Development Center, 2003). Observations are presented every decade after 1950 and less frequently before that as a way of smoothing the series.

Here's where we see properties of growth rates in action:

The growth rate of GDP/person ...

is equal to the growth rate of GDP *minus* the growth rate of “person,” a.k.a. population

Figure 3.9: Population, GDP, and Per Capita GDP for the United States



Note: Data from Maddison (2003) and the Bureau of Economic Analysis. The average annual growth rate is reported next to each data series.

Learning objectives in Chapter 4

- There are enormous differences in GDP per person, y , across countries
- The Cobb-Douglas production function $Y = A K^\alpha L^{1-\alpha}$ can help us model GDP. GDP per worker is $y = Y/L$
- This production function is realistic
 - Constant returns to scale:
If you duplicate a factory and its workers, you duplicate output
 - Diminishing returns to any one factor:
Increasing capital without increasing workers leaves idle capital
- What is technology, A ? If we assume it's the same across countries, we find a striking result: The model explains some of reality but far from all. Later, we will need to understand A

Starting Chapter 4:

Cross-country income differences are large, and we want to understand them

- Today, residents of the U.S. are:
 - 4 times richer than residents of Brazil
 - 3.5 times richer than residents of China
 - 50 times richer than those in the poorest countries
- Why?
 - We could tell many stories, but how do we assess which are more pertinent?
 - To seek an explanation, we need a **model**

What will this model look like?

- A set of several mathematical equations
- These equations have graphical expressions too, but the math is meant to be simple while providing insights and numeric predictions
- For example: the model will assert that given the inputs, GDP per capita should be \$20,000
- The model will be **highly simplified** and a clear abstraction from reality. (But that's perfectly OK)

The setup of the production model

- Suppose the economy produces only one good, which happens to be **ice cream**
- There are a certain number of workers available to produce ice cream, $\bar{L}$ (which is exogenously fixed)
- There are also a number of ice cream machines, $\bar{K}$
- Workers use ice cream machines to make ice cream according to a **production function**, $F(K, L)$

A production function

- This function takes as **inputs** the number of workers, L , and the number of machines, K
- And returns as **output** the amount of ice cream, that is produced by those workers using those machines

$$Y = F(K, L) = \bar{A}K^{1/3}L^{2/3}$$

- $\bar{A} > 0$ is a positive constant term, which we will call **productivity** — because when it is higher, output is higher even with the same K and L

$$Y = F(K, L) = \bar{A}K^{1/3}L^{2/3}$$

What is this beast, and why do we use it?!

- It is a *Cobb-Douglas* production function, and it is the most commonly used model in all of economics
- Why? It has very convenient **properties**
- The first key property is constant returns to scale:
 - If you increase all inputs by some proportion, you will increase output by the very same proportion
 - In other words, if K doubles and L doubles, then Y doubles too

The math of constant returns

$$\begin{aligned} F(2K, 2L) &= \bar{A}(2K)^{1/3}(2L)^{2/3} \\ &= 2^{1/3}2^{2/3}\bar{A}K^{1/3}L^{2/3} \\ &= 2^{1/3+2/3}\bar{A}K^{1/3}L^{2/3} \\ &= 2F(K, L) \end{aligned}$$

- Why are constant returns desirable in a model?
- If you *duplicate* your plant and workers, you would expect to *duplicate* output

How does this economy allocate its resources given the production function?

- If markets are perfectly competitive, then firms in the economy will *maximize their profits*
- What characterizes profit maximization?
- From intermediate microeconomics:
 - Firms will employ labor, L , and capital, K , up to the point at which their prices equal their marginal products
 - A key element: *diminishing marginal productivity* of K and L

What is diminishing marginal productivity of a factor like capital, K , or labor, L ?

- “Marginal productivity” means the addition to output, Y , obtained from an additional unit of the factor K or L
- “Diminishing” means it falls as K or L rises

Why does this matter? Why do we care?

- It is realistic, as we discuss on the next slide
- It plays a key role in understanding international differences in output per person

Why might marginal productivity diminish?

- Suppose our ice cream shop has **five** workers and only **one** ice cream machine
 - Workers probably bump into one another or are idle
 - One more machine probably increases output *by a lot*
- Suppose we added a machine and now have **two**
 - Workers don't bump into each other very much
 - But one more machine *now* probably increases output only some
- What if we have **five** machines and **five** workers
 - One more machine, the sixth, *might sit idle, totally unused!*
 - Output might not increase at all with the sixth machine

What is marginal productivity in this model?

- The marginal product of capital is given by the following mathematical equation (which you do not need to memorize):

$$MPK = \frac{1}{3} \bar{A} \left(\frac{L}{K} \right)^{2/3}$$

- You can see that when you hold the number of workers, L , constant, then MPK declines when K increases

Everything we have said about capital applies to labor as well

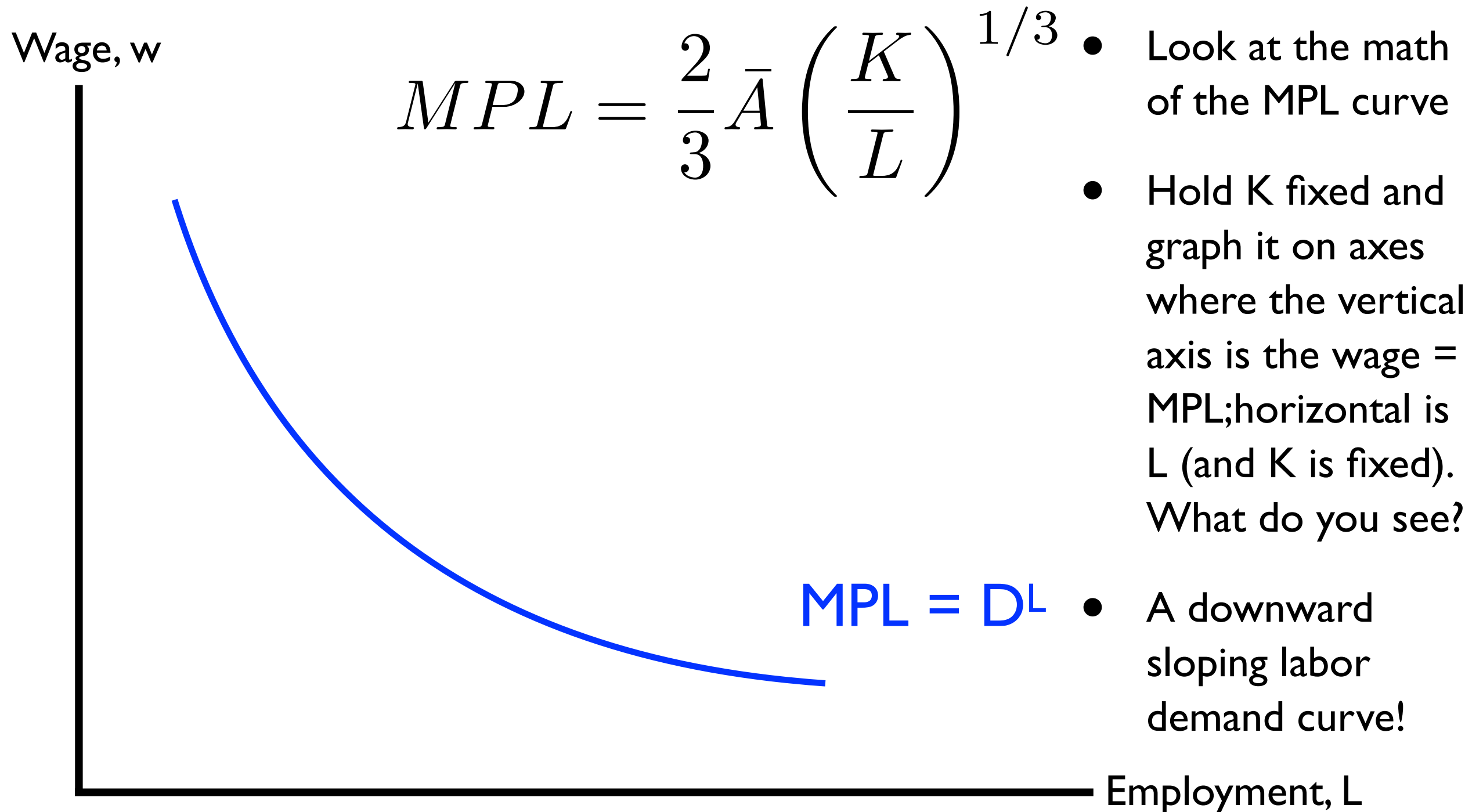
- Suppose we increased the number of workers while keeping the number of ice cream machines constant
- Each additional worker would start bumping into the other workers more and more, since there wouldn't be enough machines to go around
- The marginal product of *labor* is very similar to the MPK:

$$MPL = \frac{2}{3} \bar{A} \left(\frac{K}{L} \right)^{1/3}$$

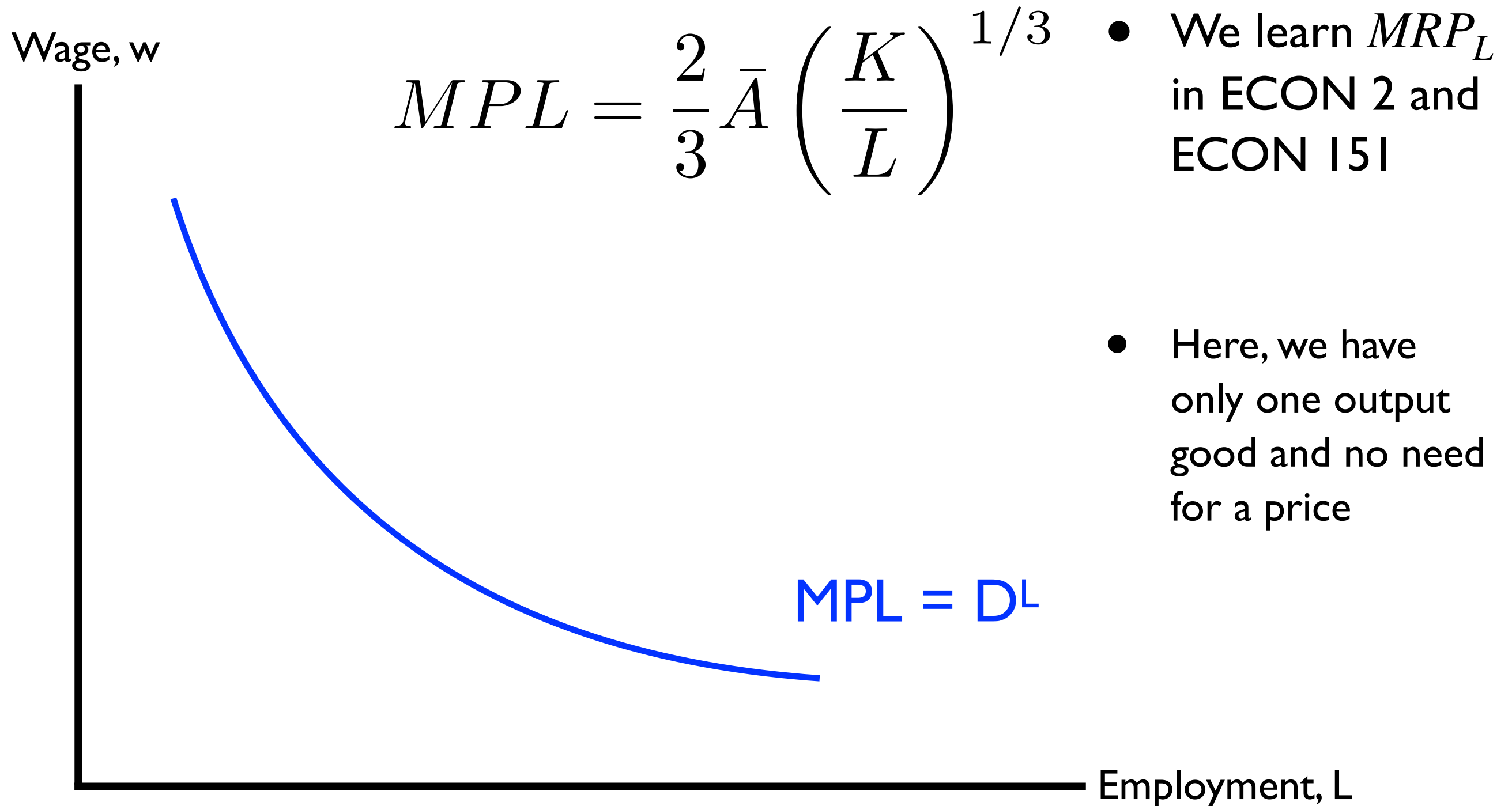
How does this economy allocate its resources given the production function?

- If markets are competitive, then firms *maximize profits*
- From intermediate microeconomics:
 - Firms will employ labor, L , and capital, K , up to the point at which their prices equal their marginal products
 - The price of labor is the **wage (w)**
the price of capital is the **interest rate (r)**
- Like always, the intersection of **supply** and **demand** determines the two prices

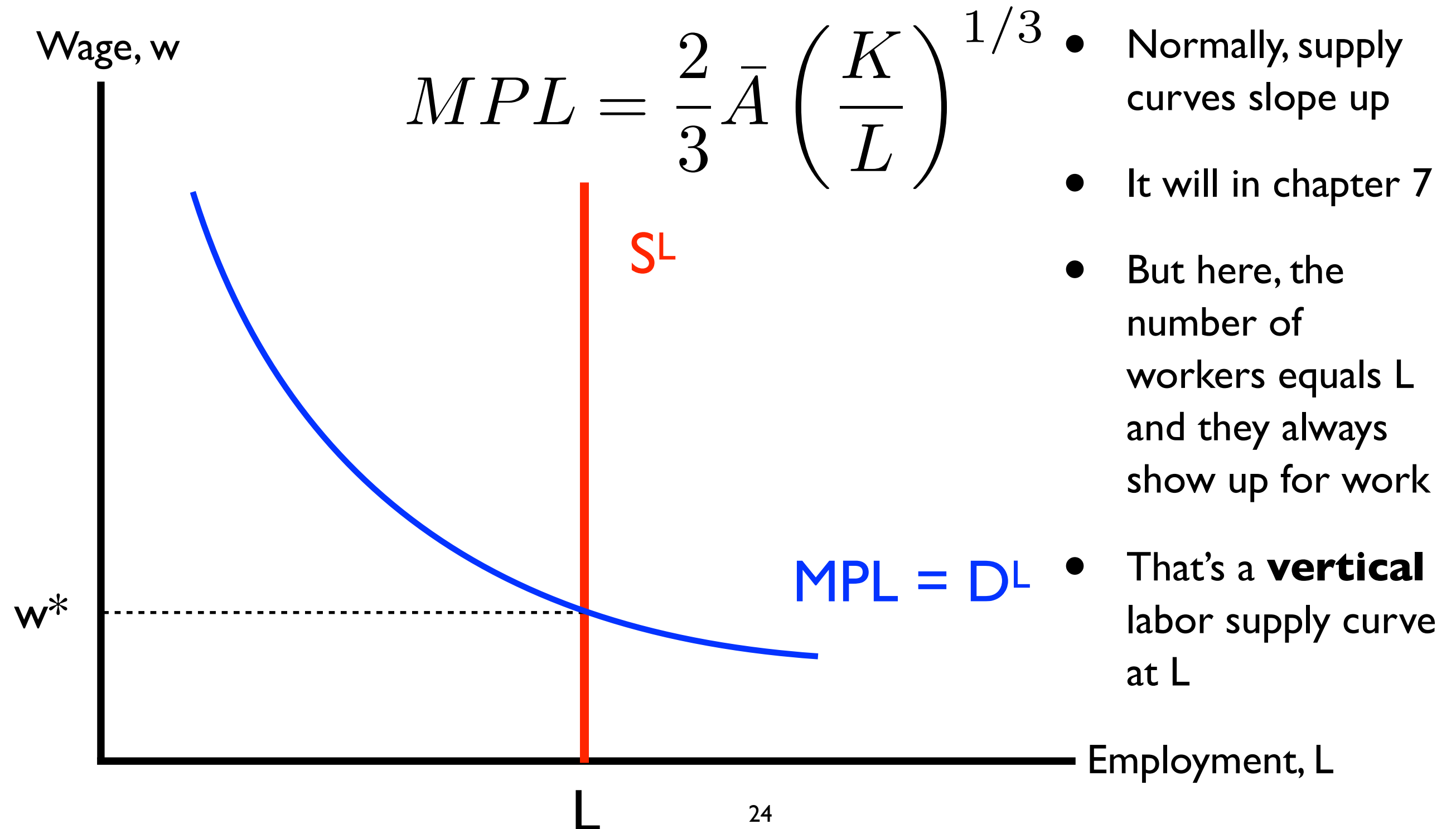
A factor's marginal product is its **demand curve**



Technical aside: The demand for labor *in a particular industry* is the **marginal revenue product** = $MRP_L = MP_L \cdot P$ with P = price of the output good



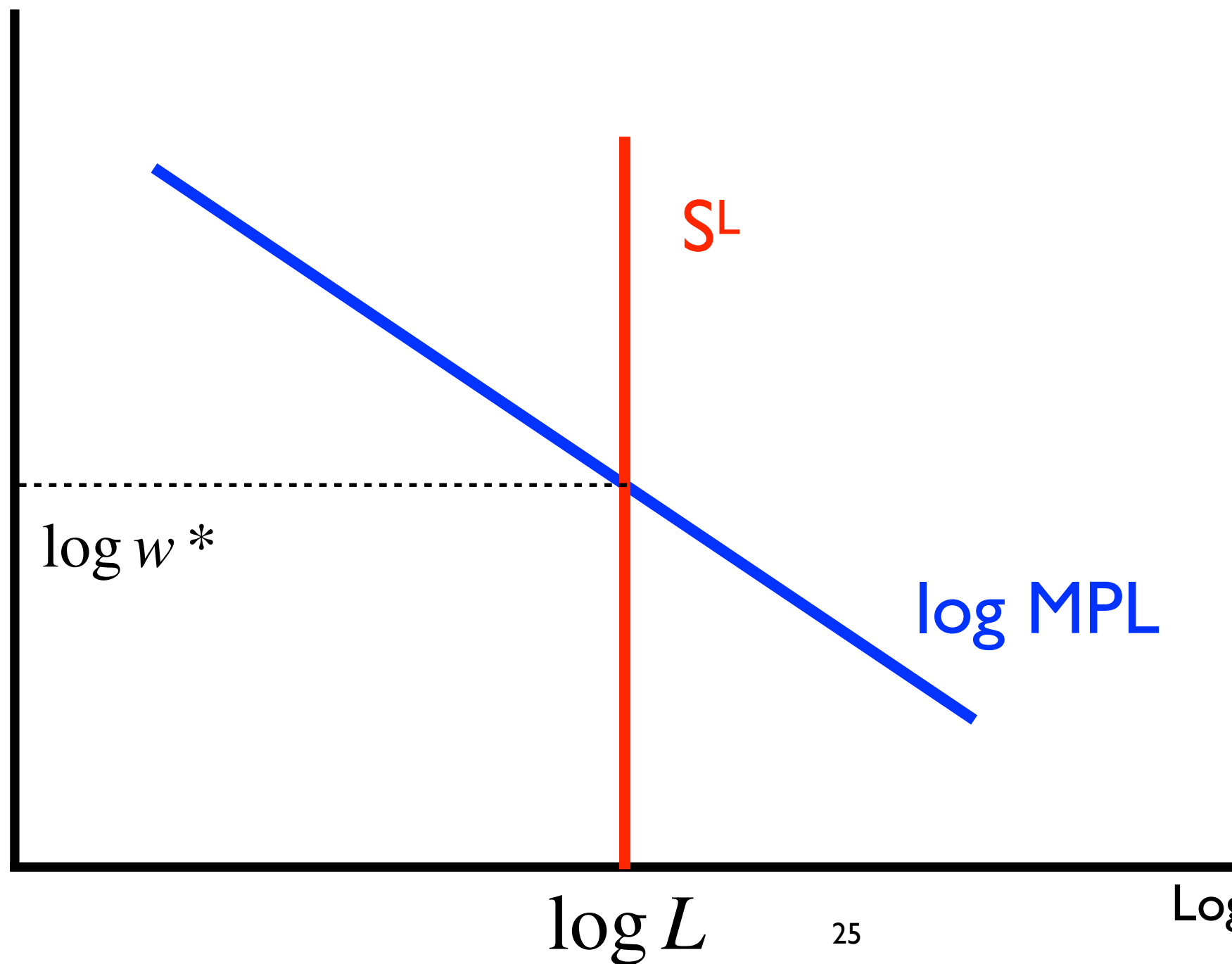
In Chapter 4, labor supply is simply the quantity of people



Suppose we took logs of both sides

$$\log w = \log \left(\frac{2}{3} \right) + \log \bar{A} + \frac{1}{3} \log K - \frac{1}{3} \log L$$

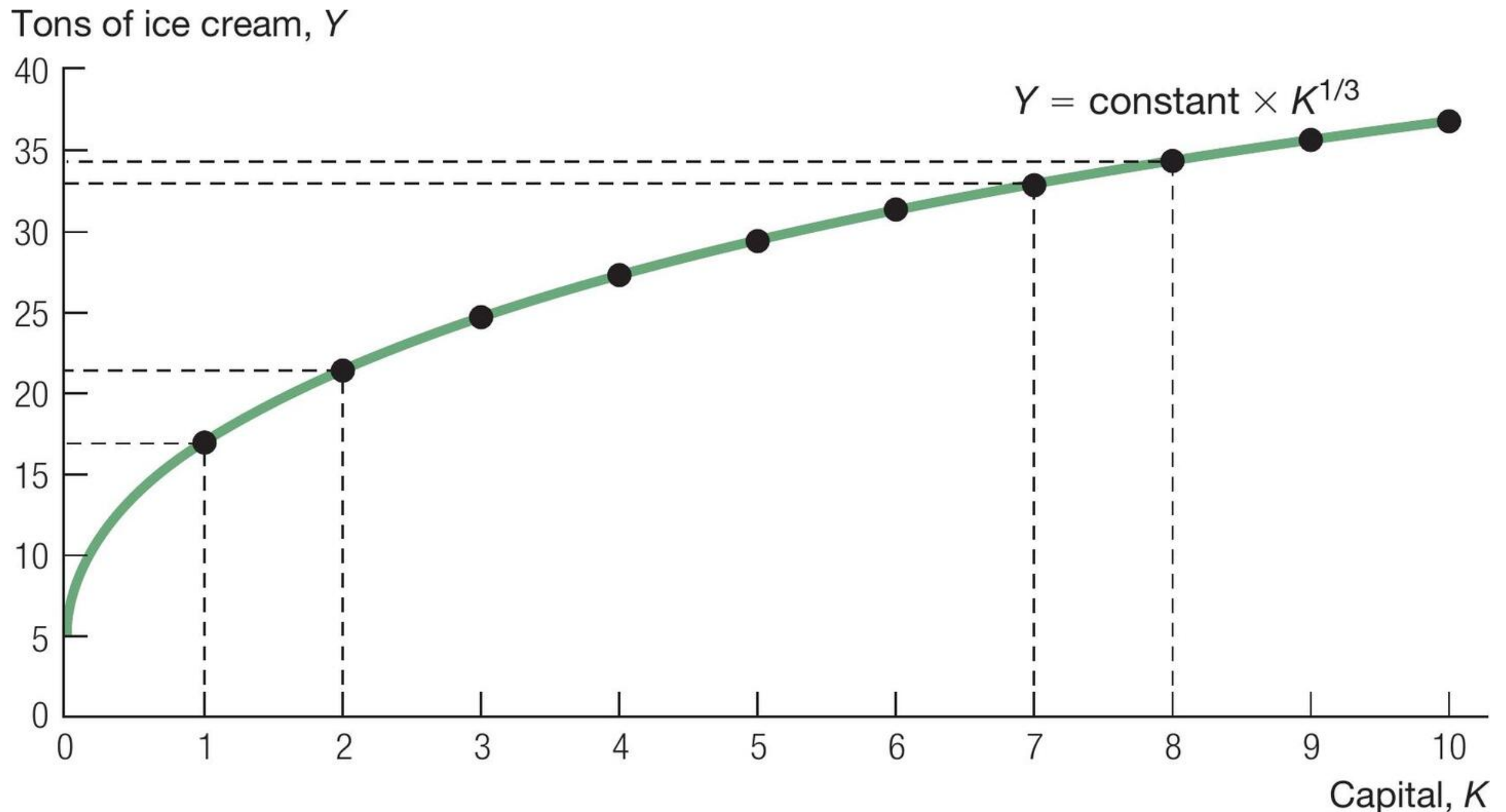
Log wage, $\log w$



- The log of a product is the sum of the logs
- What was convex is now linear
- What happens when K increases?

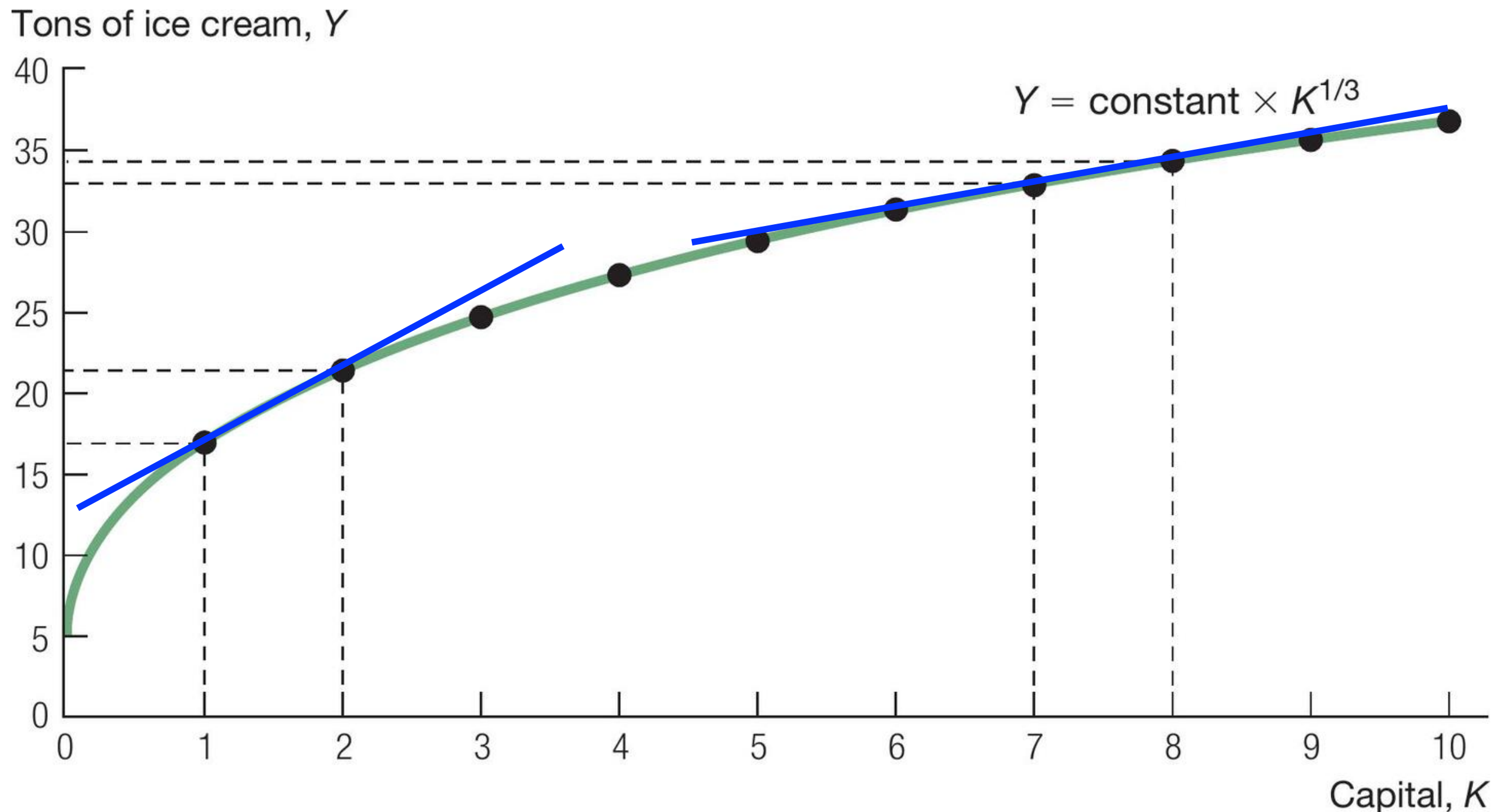
The marginal product is also the slope of a line tangent to the production function along that factor input

- Also known as the **derivative**: limit of the rise ΔY over the run ΔK as the run gets tiny



The marginal product is also the slope of a line tangent to the production function along that factor input

- Also known as the **derivative**: limit of the rise ΔY over the run ΔK as the run gets tiny



What have we learned?

1. How to set up and solve a model

- This model is simple: the supplies of K and L are constant at $\bar{K}$ and $\bar{L}$, and demand determines their prices, r and w
- We will build on this model in future chapters

2. Output, Y , is determined by capital, K , and labor, L

- More machines or workers should make more output
- We now know more about output per person

3. We understand wages, $w = MPL$, and the interest rate, $r = MPK$